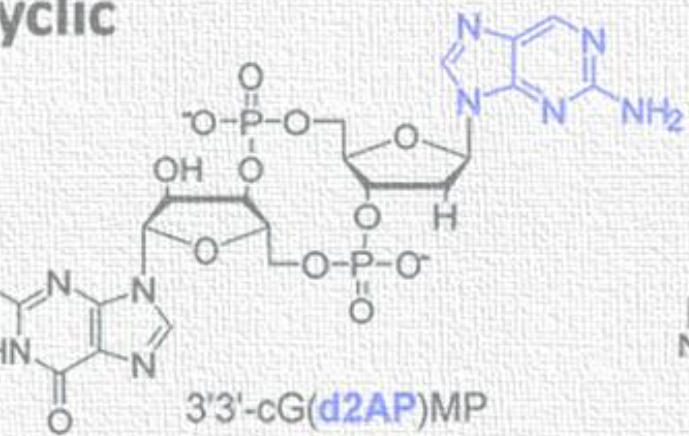
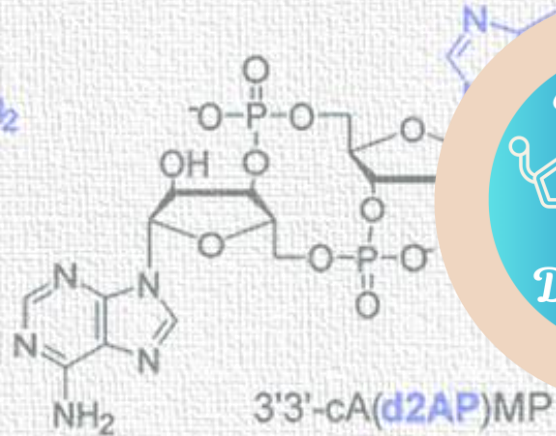


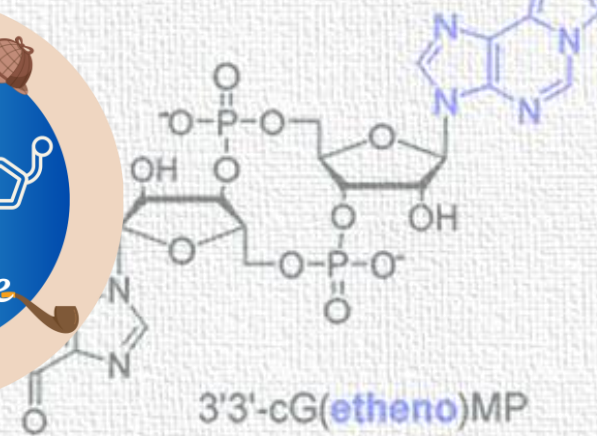
yclic



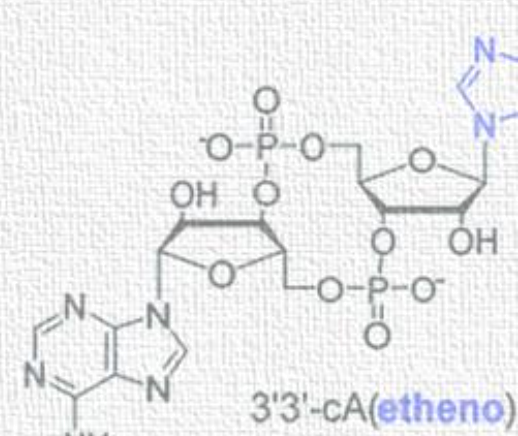
1
ex: 310 nm
em: 375 nm



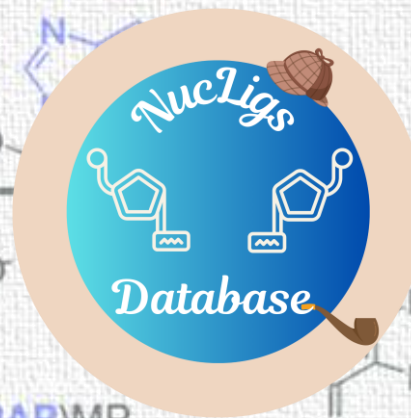
2
ex: 310 nm
em: 375 nm



3
ex: 345 nm
em: 435 nm

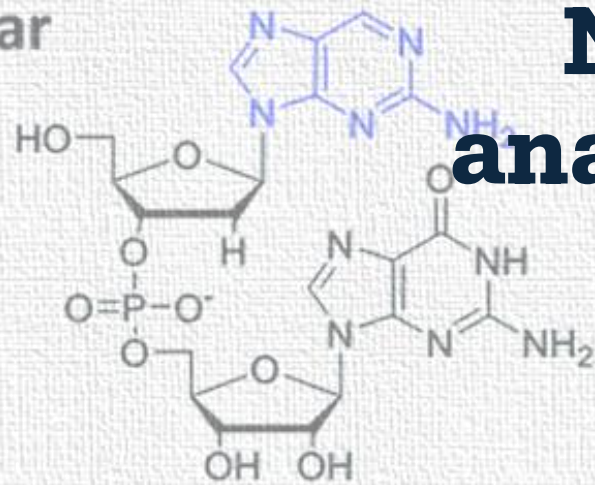


4
ex: 345 nm
em: 435 nm

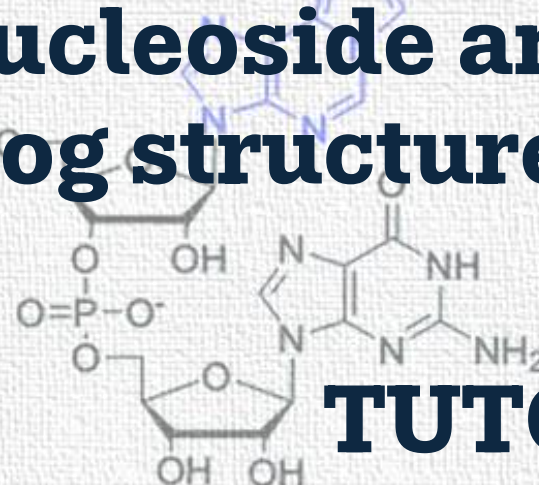


NucLigs: An online resource for Nucleoside and Nucleotide analog structure and properties

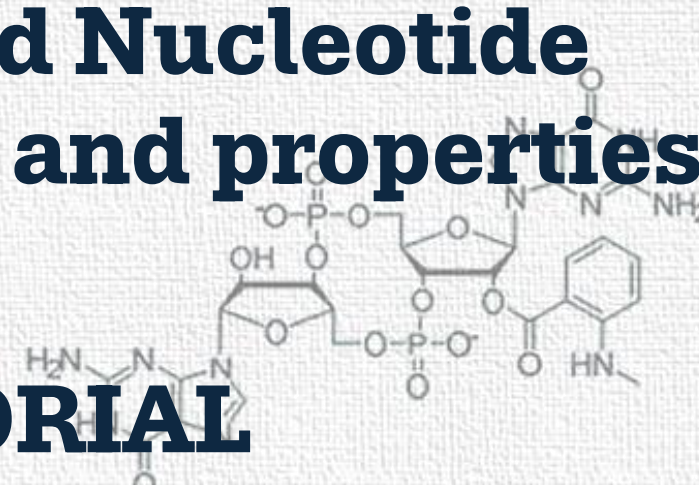
near



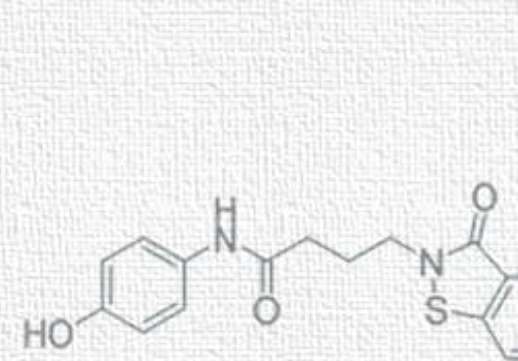
5
ex: 310 nm
em: 375 nm



6
ex: 345 nm
em: 435 nm



MANT c-di-GMP
ex: 355 nm
em: 448 nm



RocR inhibitor

TUTORIAL

1

Open and access data

Access the database using the link nucligs.streamlit.app

Click on 'Launch database' to see the dataset and related information

2

Access the dataset and its related information through the interface

3

Apply filters and sorting

Keyword-based searches

Biological targets

Therapeutic application

Mode of action

NucLigs identifier

Substructure moiety

Bibliographic filter

Entire dataset search using the left side panel

4

Extract metadata file (.xlsx format)

For single analogs 'Extract selected' button

For range of analogs 'Extract range' button

For all filtered analogs 'Extract filtered' button



5

Generate 3D structures of analogs

Click on 'Generate PDBs' button



Run cell and load the extracted metadata file for selected analog(s)

Generated_PDBs.zip

Zip file with 3D coordinate structures and a generation report



Total structure count in the database

Search bar for keywords (target, substructure moiety, NucLigs ID, mode of action, etc)

Sorting data on the basis of nucleotide bases, type of analogs and region of modification

Rearrange the displayed workbook based on different parameters and show button

Export buttons on for selected analog/ filtered analogs or a range of displayed analogs

Analog entry with respective information

The screenshot shows the NucLigs Database homepage. At the top, there's a navigation bar with a 'Tutorial' button. Below it, a header section displays statistics: 11,007 compounds, 6323 analogs, and 4658 natural. It also includes a 'Theme: Dark' selector, 'About NucLigs', 'Generate PDB', and a 'Contribute' button. A search bar is located below the header. The main content area shows a list of 11,007 results, with sorting options (Workbook order, Card View) and export buttons (Export Selected, Export Filtered, Export Range). A list of nucleotide analogs is displayed, each with its name, structure, and associated tags (Adenine, Analog, 2'-Deoxy, Sugar, etc.).

Tutorial

Tutorial accessing button

Change theme

Generate PDB structures using the exported metadata file

Understand about NucLigs database data statistics

Contribute to NucLigs database

Display panel for analog information



PubChem Database : A Nuclear Magnetic Resonance (NMR) Database

11,007 compounds

Database

an-2-yl]methoxy]phosphinic acid



STRUCTURE

RDKit.js

22 31293

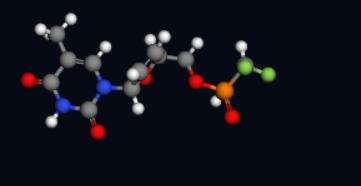
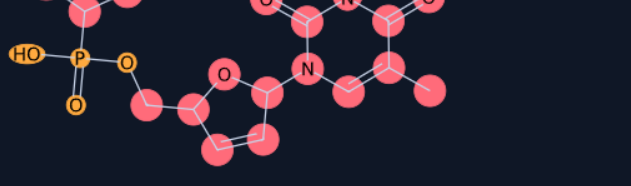
3D STRUCTURE

3Dmol.js

Stick

CPK

Surface



Fetching 3D - drag - scroll to zoom

Overview

Modification Analysis

Physicochemical

Biological Targets

Bioactivity

Clinical & Regulatory

References

Identifiers & SMILES

QUICK FACTS

STRUCTURE IDENTIFIER
Nucl_085147

TRADE / COMMON NAME
difluoromethyl-[(5S)-5-methyl-2,4-dioxypyrimidin-1-yl]-2,5-dihydrofuran-2-yl[methoxy]phosphinic acid

TYPE
Analog

MOLECULAR FORMULA
C11H13F2N2O6P

MOLECULAR WEIGHT
338.20 g/mol

LOGP
0.723

CHEMBL ID
CHEMBL1126397

PUBCHEM CID
44283173

NUCLEOSIDE SCAFFOLD

MODIFICATION GROUP (1-M)

BASE (COL 1)
Thymine
Pyrimidine - 5-methyl-2,4-dioxypyrimidine

SUGAR (COL 2)
-
Phosphate: Mono

MODIFICATION (COL 1-M)
Fluoro

TARGET (W-AM)

COMPOUND B TARGET (COL W)

Human immunodeficiency virus type 1 reverse transcriptase

BIOACTIVITY (COLS A-D-Q)

BIOLOGICAL ACTIVITY DATA

IC₅₀
300.00 ugaL

NucLigs_SMILES_to_3D.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

Show code

NucLigs Database • Structure Generator

NucLigs 3D Structure Generator

Generate Desired 3D Structures in PDB Format

This professional module is a part of the **NucLigs Database** and enables automated generation of optimized **3D molecular structures in PDB format** from user-provided SMILES, CSV, or Excel files. Designed for computational biology, cheminformatics, RNA–ligand studies, docking, molecular dynamics, and structure-based drug discovery workflows. User need to upload the exported file from the NucLigs database in the cell and the tool will automatically generate the ZIP folder with all PDB format structures and an additional CSV report.

INPUT MODES
SMILES / CSV / XLSX

OUTPUT
Optimized 3D PDB

Generate 3D structures from resulted file

[1] Show code

37.1/37.1 MB 24.9 MB/s eta 0:00:00
4.9/4.9 MB 68.4 MB/s eta 0:00:00

Paste SMILES Upload File

Aspirin, CC(=O)OC1=CC=CC=C1C(=O)O
Caffeine, CN1C=NC2=C1C(=O)N(C(=O)N2C)C

OR

CCO
CCN

Paste SMILES → Click 'Generate PDBs'

Generate PDBs

User may either paste SMILES OR uploaded the extracted metadata file to generate 3D coordinate structures of desired analogs

Generate 3D structures from resulted file

[1] Show code

37.1/37.1 MB 24.9 MB/s eta 0:00:00
4.9/4.9 MB 68.4 MB/s eta 0:00:00

Paste SMILES Upload File

Upload CSV/XLSX

Automatically detects Name/NL and SMILES columns. Ignores other columns.

Upload (0)

Generate from File Delete File

Click on 'Upload'

Select the exported metadata file

Click on 'Generate from File'

